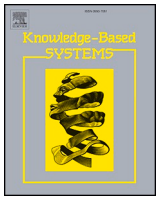
Knowledge-Based Systems 283 (2024) 111123



Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys





A machine learning framework for quantifying in-game space-control efficiency in football

Chaoyi Gu [a,*], Varuna De Silva [a], Mike Caine [b]

[a] Institute for Digital Technologies, Loughborough University, London, United Kingdom
[b] Aston University, Aston, Birmingham B4 7ET, United Kingdom

ARTICLE INFO

Keywords:
Machine learning
Deep generative model
Football
Performance analysis
Space-control

ABSTRACT

Analysis of player tracking and event data in football matches is used by the coaching staff to evaluate team performance and to inform tactical decision-making, whereas using Machine Learning methods to gain useful insights from the data is still an open research question. The objective of our research is to discover the football team's space-control efficiency using a novel Machine Learning approach and evaluate the team performance based on its space-control efficiency. We develop a novel Possession Evaluation Model through deep generative machine learning to predict the football team's space-control capability utilising tracking and event data. The developed model is used to quantify the efficiency of attacking and defending for a given sequence of play. Performance analysis results demonstrate that this novel method of space-control efficiency quantification is objective and precise. The superior performance of the model is attributed to the utilization of deep generative modelling on image datasets and conditioning in the prediction with contextual factors. This study presents a novel approach to football analysis in evaluating team performance and providing tactical insights for the coach to make data-informed adjustments.

1. Introduction

Football team performance analysis is complex as there are many variables and factors which in aggregate determine the outcome of the game [1]. In previous studies, on-pitch factors like players' fatigue and team spirit and off-pitch factors such as weather conditions are considered for predicting the outcome of the game [2–4]. There are two limitations of these studies. Firstly, the definition and the calculation of many factors used come from the human definition, which can be error-prone and make the model unreliable. Secondly, the lack of detailed contextualised data in football games makes the model hard to be applied by the coaching staff to analyse the game. Besides, in one previous study, the passes were used to frame passing networks which are then used to predict the game outcome. Only using passing as the factor without considering other events like dribbling makes the results of predicting the chance of success in attacking unreliable [5]. To analyse the football game thoroughly and objectively, the contextual factors considered are the key. When one team loses a match with a much higher possession rate and more shots, it is hard to judge if the cause of the failure is poor team performance [6]. Therefore, evaluating the game only on the basic statistics provides only partial insights. To make more objective and accurate evaluations, current studies mainly use statistical methods to analyse football match data. For example, the 'goal probability model' is built to find the factors deciding the scoring possibility and the 'pass feasibility model' is used for calculating the successful rate of passes given contextual information [7,8]. These statistical approaches can be useful for deriving connections between variables from the sample data. However, these approaches are unable to find generalisable predictive patterns which are crucial to answer what-if questions when analysing football matches. For instance, to analyse the efficiency in defending, Stein et al. conduct research to quantify space controlled by the attacking team and evaluate the defending efficiency based on the changes of the quantified controlling space from the attacking side [9]. When quantifying the controlled space, they only analyse the context near the ball trajectory without considering other players' behaviours. Subsequently, their evaluation of defending performance can be biased since they fail to answer the what-if question of whether the team will sacrifice even more space elsewhere on the pitch when covering the space near the ball carrier.

* Corresponding author.
E-mail address: c.gu@lboro.ac.uk (C. Gu).

https://doi.org/10.1016/j.knosys.2023.111123
Received 29 March 2023; Received in revised form 8 October 2023; Accepted 28 October 2023
Available online 13 November 2023
0950-7051/© 2023 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

1.1. Related work

To evaluate team performance in attacking, Fernández and Bornn proposed a statistical technique to measure space creation and generation in professional football games [10]. They presented a method to quantify spatial value occupation and generation in active plays of the games. The space creation denoted the space created by a player for himself/herself and for opening up space for teammates by attracting opponents to leave their original positions. To construct these two space value metrics, they introduced a spatiotemporal framework including two sub-models. The first was the Pitch Control Model which calculates the possibility of the attacking team controlling the ball at a given location of the pitch based on motion information, player position and relative distance to the ball. The second was a model to quantify the value of the given location on the pitch based on the ball position. With space generation and space occupation metrics, they evaluated players' contribution to attacking. There are two limitations of this study. At first, with their metrics, it is applicable to evaluate individual player's performance, but it is hard to apply the metrics in analysing the overall team performance. Secondly, this research is only used for analysing attacking behaviours without providing insights for the defensive side. Moreover, in their analysis, they directly used their metrics to assess players' performances without any benchmark for comparison. Accordingly, it was hard to tell what metric score suggested a good or bad performance. Another problem is that they didn't condition the performances of players on their different roles, which could lead to bias when making evaluations. Another study looked into passing within attacking scenarios specifically. The research proposed a framework to assess passing effectiveness in professional football [11]. The framework contained two parts, the number of defenders between the goal and ball carrier, and the attacking space dominance. The attacking space dominance was quantified by implementing Voronoi diagrams based on a tessellation of the football pitch into cells according to the positions of the players [12]. They applied Voronoi diagrams to calculate the space controlled by the attacking team for a given game state and used the difference between the area controlled before and after the pass to evaluate the passing effectiveness. The limitation of this study is that for a given game state, the evaluation of passing effectiveness is only based on their proposed statistical calculation with no benchmarks to compare. For a specific pass, what other players will do in the same context is unknown and it is hard to judge the pass objectively without comparing it to others.

To evaluate team performance in defending, Andrienko et al. conducted research to detect and quantify the defensive pressure in football games and built an interactive tool to visualise the defensive pressure [13]. They quantified the pressure with a statistical method which related the pressure to the directions from the target which was the ball carrier or the ball to the defended goal and to the teammates of the ball carrier. With this quantification, they proposed an interactive tool to show pressing heatmaps for a given game state, which was expected to help coaching staff make tactical analysis. This research can be useful as a visualisation tool in enabling coaching staff to analyse team-pressing behaviour more conveniently. However, this research can hardly provide evident insights about the pressing in football games since the quantification of defensive pressure in this study is subjective and can be misleading without justification from experts. For instance, when the defending team presses the opponent too heavily upfront on the pitch and fails to cover the space behind, the pressing can still be evaluated as good by their quantification method. This research, therefore, could be inefficient in defending tactical analysis since the results will be validated only when embedding human expertise. In another research conducted by Stein et al., team's defending behaviours are analysed by quantifying space controlled by attacking teams [13]. They only selected game states before shooting to do analysis. They identified opposing players who were close to the ball trajectory to be candidates responsible for controlling the free space around the ball carrier or receiver and calculated the difference between space expected to be covered and space actually covered. Then they evaluated the team's defending behaviours based on each identified candidate player's performance. There are several problems and limitations in this study. At first, the identification of candidates who are supposed to cover the open space is subjective since the distance between the candidate to the ball trajectory is defined manually. Secondly, they only analysed the context near the ball trajectory when evaluating the defending behaviours without considering other players' behaviours. Specifically, if the candidate moves as suggested in their research to cover the free space around the ball carrier, the team will be potentially at risk of losing more open space which was previously covered by this candidate. This trade-off was not calculated in their research and only the partial context is considered when analysing the team's defending behaviour, which could lead to misleading results. Specifically, when the proposed model in this research suggested the team did well in covering space, the actual performance of the team could be bad since the sacrificed originally covered space by the candidate player was not calculated in their model. The limitation of this research is that the overall analysis of the whole team is missing while only each individual is looked into.

Current studies on football team performance analysis have two main limitations. Firstly, the generalizable patterns in football matches are not well researched since most studies use statistical methods and only manage to discover the direct relationships between factors influencing the football game without finding the hidden patterns. Some studies aim to dig deep into the data embedding the domain knowledge when making statistical analysis, which can limit the findings from the data and lead to unobjective results. Secondly, it is hard to apply these studies when aiming to objectively analyse the game since there is no benchmark provided when analysing a specific game state [10,11]. To be more specific, what other teams will do in the same game state is unknown when evaluating one team's performance. As a result, the coaching staff can hardly gain useful insights mainly based on these studies.

1.2. Aim and objectives

Our research narrows the gap between current research and the solution to the main research question which is to objectively and efficiently analyse football team performance by training a generative

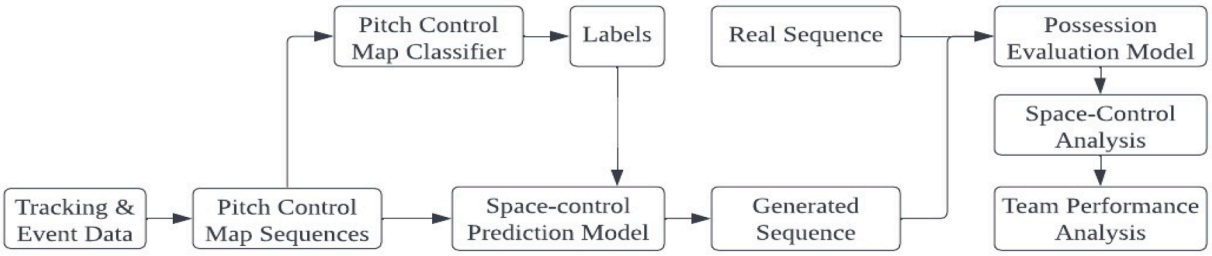


Fig. 1. The overall framework for quantifying team space-control efficiency and evaluating team performance.

model which can make generalizable predictions in team performance for a given game state and using it as a benchmark to quantify the team space-control efficiency and evaluate specific team's performance accordingly and objectively.

In general, there are four major contributions of our research:

1. We quantify the team space-control efficiency in given in-game possession and this quantification can be used for further team performance analysis.
2. We provide an end-to-end pipeline to evaluate the team performance given raw match data. When analysing previous games, the coach can quickly locate the possessions deviating from the expectations.
3. After locating the anomalous possession, the coach can make analysis and adjustments based on the results of our Possession Evaluation Model.
4. Coaching staff can evaluate individuals' contribution to team space creation utilising our framework.

2. Methodology

To achieve the research objective, which is to evaluate the football team's performance by quantifying the efficiency of controlling open spaces on the pitch, we propose the framework shown in Fig. 1. The tracking and event data are converted to Pitch Control map sequences, with every sequence corresponding to one possession within the football match at first. Pitch Control map quantifies the space controlled by each team at a specific time-step in the football game [14].

After the tracking and event data are converted to Pitch Control map sequences, the Space-control Prediction Model is trained on these Pitch Control map sequences with a pre-trained Pitch Control map Classifier embedded to condition the model prediction on the label of each frame. The Space-control Model can generate the whole sequence when only given the first frame and labels for the following frames. The generated sequence represents league-average performance. Then, we propose the Possession Evaluation Model to evaluate the team's efficiency in controlling the valuable space on the pitch by comparing the sequence of Pitch Control maps for real life and the sequence generated by the Space-control Prediction Model using EPV (Expected Possession Value). EPV quantifies the value of possessing the ball when given the ball position and match state, such as open play or set piece [15].

Lastly, we use the Possession Evaluation Model to quantify the team space-control efficiency in football games and evaluate team performance based on the quantification.

2.1. Dataset

Tracking data and event data of 43 matches in 2019–2020 Premier League Season with all twenty teams participating in, are used in this study. Data are cleansed and pre-processed by remaining only active plays while the set-pieces, penalties, and other possessions when the game is interrupted are removed from the dataset. Considering that data in professional football is highly confidential, we only have limited access to it without having data from the whole season for training the model. Despite that, we still have a substantial amount of data to train our models and achieve good results.

2.2. Data pre-processing

The tracking data and event data are used to generate Pitch Control maps. Each possession within the game is represented by a sequence of Pitch Control maps with lengths varying from 2 to 110 and 1 s as the timespan between each two adjacent frames.

10 % dataset, 20,000 sequences in total, are selected for training Pitch Control Map Classifier. The data is split into training and validating datasets, with 80 % and 20 % of the whole dataset for each. Since the classifier is used to classify the connection between every two adjacent frames, the length of the input sequence should be 2. Longer sequences are processed by sliding window to generate the input with the required length. After the processing, there are 52,000 sequences in the training dataset and 13,000 sequences in the testing dataset. Subsequently, we manually label all the sequences as pushing, backing, or unlabelled.

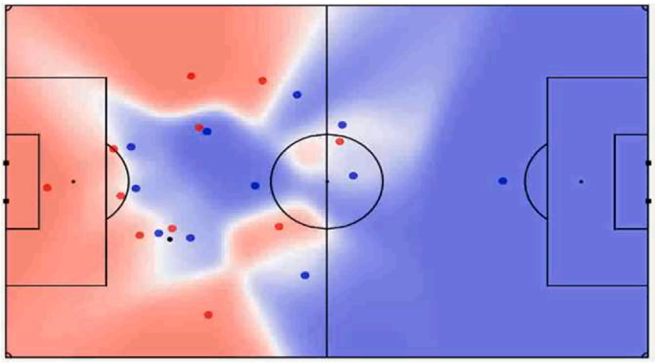


Fig. 2. Pitch Control map.

The other 90 % of the dataset, 180,000 sequences in total, are filtered with those having lengths no less than 6 remaining. All the sequences are split into training and validating datasets with 80 % and 20 % of the whole dataset. The sequences longer than 6 are processed by sliding window to generate the input with lengths as 6. After the processing, there are 400,000 sequences in the training dataset and 100,000 sequences in the testing dataset. The datasets are used for training the Space-control Prediction Model.

2.3. Pitch Control map classifier

As shown in Fig. 2, the Pitch Control map quantifies the region controlled by each team at a specific time-step in the football game [14]. There are two steps to get the Pitch Control map. The first step is training a pass probability model, which can predict the successful rate of a pass when given all players' and the ball's positions, speed and facing directions, and the player ID of the pass receiver as input. As shown in Eq. (1), each pass is treated as a Bernoulli trial and k represents the outcome of the pass with success = 1 and failure = 0. Whether the pass will be successful or not is derived from the input x and two trained parameters σ and λ.

$$p(k|\sigma,\ \lambda, x) = \begin{cases} 1-p & for\ k=0 \\ p & for\ k=1 \end{cases} \tag{1}$$

The pass probability model is trained to find the set of parameters that maximise the likelihood shown in Eq. (2).

$$\mathscr{L}(\sigma,\ \lambda|k, x) = P(k|\sigma,\ \lambda, x) \tag{2}$$

The second step is using this trained pass probability model to compute the stationary pass probability for an imaginary ball placed at a given location on the pitch and convert the whole pitch to a real-valued scalar field called Pitch Control Function for generating a Pitch Control map.

Pitch Control map can be used for analysing football game scenarios since the visualization of controlled area by each team can help coaches evaluate the attacking and defending tactics [16].

Moreover, utilising open space is the key to creating more scoring chances in a football game [10]. To understand further the connection between open areas and team tactics, utilising the Pitch Control map can be an ideal option. Besides, with the Pitch Control Function, the high-dimensional football data can be processed into images, which enables more options to analyse the game using Machine Learning techniques such as Variational Autoencoders.

To condition the reconstructed sequence on the label of each frame in

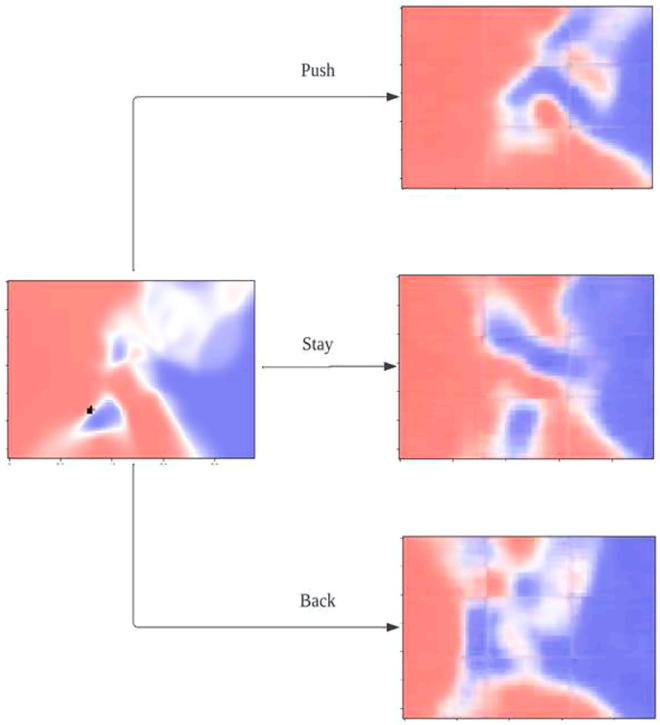


Fig. 3. Visualization of three different types of Pitch Control map sequences.

training and generation, we need to label the Pitch Control maps before training the Space-control Prediction Model. We train a Pitch Control map Classifier on manual labelled Pitch Control map sequences with a length of 2. As shown in Fig. 3, each sequence is labelled as push or back, which indicates the movement of controlled space by the attacking team on the pitch. For two frames which have similar patterns, it is hard to label. These sequences are left as unlabelled. All labelled samples are used for training the classifier. In prediction, the sequences with confidence less than 0.95 are those that can hardly be identified as push and back. These sequences don't show detectable pushing or backing patterns, so they are classified as staying.

Moreover, there are two reasons why we use 10 per cent of the whole dataset to pre-train a classifier to generate labels for frames instead of directly manually labelling the whole dataset. At first, the whole dataset is too big to label manually. Secondly, when two adjacent frames are similar, it is hard to label the sequence correctly. With this classifier, we can label sequences with higher validity [17].

We use CNN-LSTM framework combining Convolutional Neural Network (CNN) and Long Short-Term Memory Network (LSTM) to build the classifier.

Convolutional Neural Network (CNN) is a deep neural network architecture widely used in image processing [18]. A CNN consists of several convolutional layers shifting windows across the input with nodes. These nodes share weights and abstract the input to feature maps which can be further used for different research needs, such as image classification tasks.

Long Short-Term Memory Network (LSTM) is a neural network architecture used for time series problems [19]. Compared to the Recurrent Neural Network (RNN), it can do much better in memorising long sequences with a built-in forget gate, an input gate, a new candidate gate and an output gate [20].

From the architecture of the classifier depicted in Fig. 4, the frames at t and t-1 are inputs which go through the same encoder at first. The encoder is a group of neural networks consisting of Convolutional Layers and Fully Connected Layers. After going through encoders, the two vectors containing extracted information from two input frames, are down sample to vectors and become the input of LSTM neural networks. Subsequently, the two output vectors of LSTM are concatenated to one vector and go through a down sampling dense layer to a vector with the same length as that of the label vector. The classifier is trained by minimising the Cross Entropy loss between the predicted label and the real label.

2.4. Space control prediction model

To predict the possession when given only the first frame, we build a generative model which can predict the sequence conditioned on the first frame and the condition for the frame at each times-step. We build this model on top of the structure of VRNN (Variational Recurrent Neural Network).

Variational Recurrent Neural Network (VRNN) is an extension of Variational Autoencoder (VAE). VAE, as one type of generative model, is used to generate new samples from learning hidden representations of input data in terms of regularities and patterns. VAE is composed of encoder, latent space and decoder [21]. The main idea is to extract key features of inputs, such as images, by encoding, to low dimensional latent space. Compared with autoencoder, which has similar model architectures with encoder and decoder embedded, VAE is much more flexible and, in most cases, able to overcome overfitting so that it can generate more reasonable and stable new samples when dealing with complicated input data [22]. This edge comes from the difference in latent space representations. VAE uses distribution rather than fixed

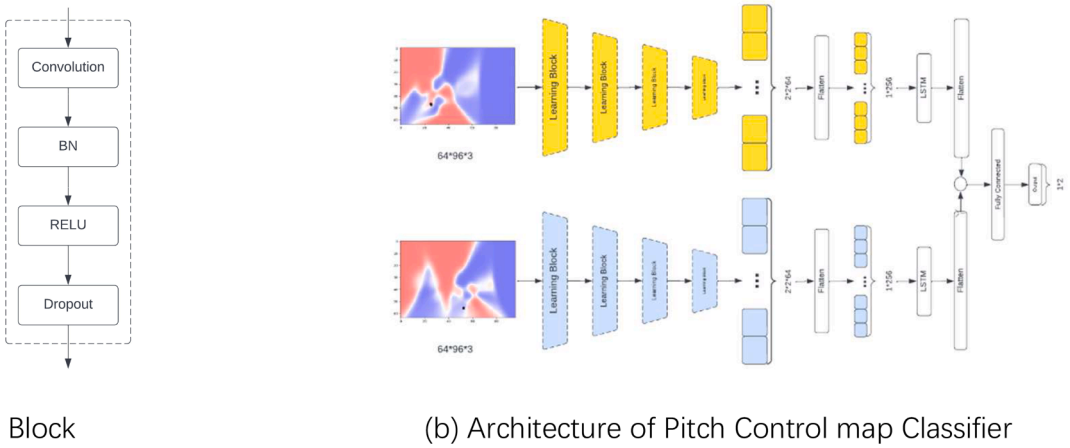


Fig. 4. (a) Block diagram of the local feature learning block. Batch Normalization and Rectified Linear Unit are abbreviated as BN and RELU. (b) Architecture of Pitch Control map Classifier which is used to label Pitch Control maps.

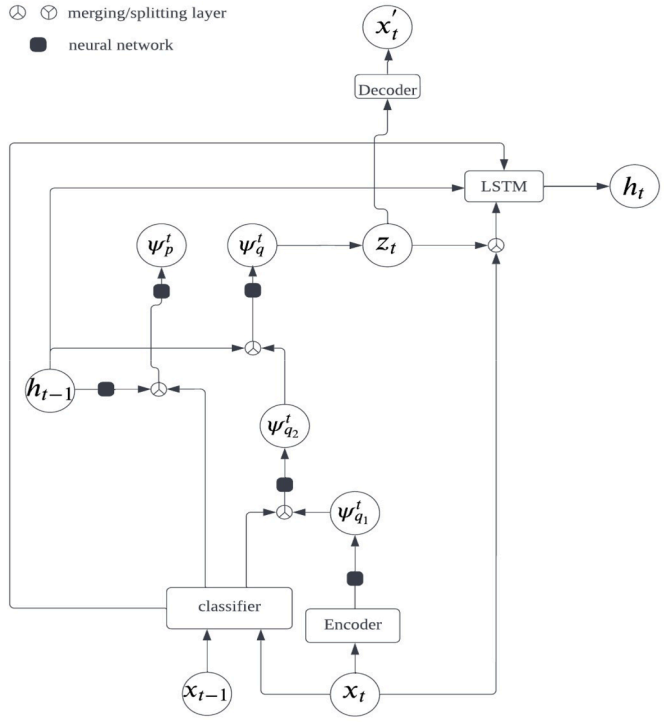


Fig. 5. Architecture of Conditional VRNN Model.

vectors to represent the latent space. To do backpropagation, which is crucial to training, the reparameterisation trick is invented to convert the latent space distribution to a normal distribution parameterised by the mean and log-variance vectors. After that, the normal distribution will be queried by random sampling and calculated by the parameters, which then go through the decoding process and get reconstructed. The loss function consists of KL divergence (the difference between the latent distribution and the standard Gaussian distribution) and reconstruction loss (the difference between the reconstructed images and inputs) that could be mean squared error, cross-entropy or others based on the input data structure.

VRNN combines VAE with a recurrent neural network to model sequential data [23]. It explicitly models the dependencies between latent random variables across subsequent timesteps. The VRNN contains a VAE framework at every timestep, and the parameters of each VAE are conditioned on the state variable of a recurrent neural network. This addition enables the VAE to take the temporal structure of sequential data into consideration when getting trained.

Like standard VAE, the generative and inference models are learnt jointly by maximising the variational lower bound. Besides, the objective function is a timestep-wise variational lower bound since the input of the model is sequences rather than individuals.

VRNN Model (also known as the Vanilla VRNN Model in machine learning) can generate sequences by randomly sampling the latent space. However, the randomly generated frames cannot be controlled and used for meaningful inference. Therefore, we propose a novel Conditional VRNN architecture with a pre-trained Pitch Control map Classifier embedded to generate the sequence conditioning on the same labels from the real-life sequence. The generated sequence represents the likely team performance in this possession.

In the following subsections, we explain how we construct the Space Control Prediction model using Conditional VRNN.

2.4.1. Model architecture

The overall structure of the model is shown in Fig. 5.

X_t denotes the pitch frame at time t, h_t denotes the hidden state of LSTM networks at time t and z_t represents the latent space at time t. We use ⋮ to represent the merging layers, ⋮ to represent the splitting layers, and ■ to represent the neural networks.

2.4.2. Frame encoder

The encoder is built with stride convolutions and Rectified Linear Units (ReLu), which extract latent representations from the input frame at each timestep of every sequence. The encoder can be used to lower the dimension of the input, Pitch Control Maps.

2.4.3. Prior and approximate posterior

The prior at time t is ψ_p^t which is parametrised as a diagonal normal distribution $\mathcal{N}(\mu_{0,t}, diag(\sigma_{0,t}^2))$ where $\mu_{0,t}$ and $\sigma_{0,t}$ are extracted from h_{t-1}, the recurrent hidden state at time t-1. The approximate posterior at time t is ψ_q^t which is parametrised as another diagonal normal distribution $\mathcal{N}(\mu_{z,t}, diag(\sigma_{z,t}^2))$ where $\mu_{z,t}$ and $\sigma_{z,t}$ are extracted from the concatenation of latent variables encoded from input frames and h_{t-1}. The posterior at time t will be used to infer getting z_t which can be used to reconstruct the input image.

The prior represents the probability distribution with an uncertainty of the Pitch Control Maps before observing the input, while the posterior represents the probability distribution conditioned on the observed input Pitch Control Maps.

2.4.4. Frame decoder

At each timestep t, the decoder takes the representation of the previously generated frame x_{t-1} and variables z_t sampled from the posterior to generate x_t based on $p(x_t|z_t, x_{<t}.c)$. The decoder consists of transposed convolutions and Rectified Linear Units (ReLu).

The decoder is used to reconstruct the Pitch Control Maps.

2.4.5. Recurrence

To capture the temporal information amongst frames within a sequence of Pitch Control Maps, the LSTM network is applied to denote the recurrent relationship amongst frames and help generate consistent sequences. The hidden state h_t at time t is updated according to the encoded input frame x_t, latent variables z_t, reshaped conditional variables c_t and hidden state h_{t-1} at the previous timestep. The initial hidden state is set to zero vector.

Since our input is a sequence of Pitch Control Maps, utilising the LSTM network can help the model capture the temporal connections amongst frames, which is crucial for the model to make reliable predictions.

2.4.6. Conditioning

The conditions are acquired by feeding the two contiguous frames to a pre-trained classifier that outputs the vector $[P_1, P_2]$. P_1 and P_2 representing the possibility of the short sequence being pushing and backing. While P_1 and P_2 are both less than 0.9, the prediction of the label is viewed as ambiguous, and the sequence is labelled as staying. Then the generated labels are processed by one-hot encoding before being implemented into VRNN. Three approaches are made to implement conditions into the VRNN Model. The technical details are shown in the appendices.

The conditioning is added to help the model capture the patterns between every two adjacent Pitch Control Maps of each sequence, which is important for the model to make predictions closer to the real-life sequence.

2.4.7. Model learning

The loss function contains the difference between posterior and prior (KL divergence) and reconstruction loss (Structural Similarity Index Measure). The loss is calculated at each frame and then the mean of losses across all frames is calculated. Detailed functions are shown in the Appendices.

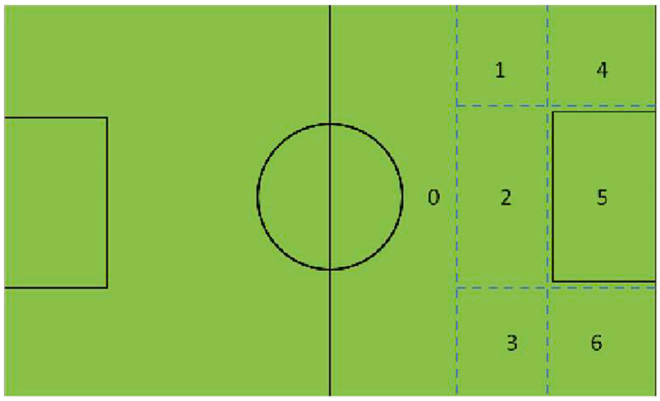


Fig. 6. Zones on the pitch [11].

2.4.8. Prediction

When making predictions of space-control, we use the first Pitch Control Map of given possession as the input. Then, the input goes through the encoder and becomes the latent space vector which is the lower-dimensional extraction of the input frame. With recurrence and conditioning, the latent vectors corresponding to the future Pitch Control Maps can be predicted. In the last, each latent vector will go through the decoder for reconstruction. The reconstructed frames represent the predicted Pitch Control Maps.

2.5. Possession evaluation model

With the trained Space-control Prediction Model, we can generate the whole sequence of Pitch Control maps when given the first frame and the conditions for each time-step. Since our model is trained on possessions in Premier League matches, each frame generated by our model is thought to represent the expected performance. For instance, when given a previous frame and condition as pushing, the model can generate what is most likely to happen on the pitch when a team pushes forward. Therefore, for each possession of a specific team, the first frame and all the conditions between every two adjacent frames are extracted and used to generate a sequence of Pitch Control maps showing the most likely changes on the pitch when other teams are in the same situation. Expected Possession Value (EPV) is used as the metric for comparing real possession sequences and generated sequences.

Expected Possession Value was first introduced in 2011 [15]. As shown in Eq. (3), it quantifies the value of possessing the ball when given the ball position and match state such as open play or set piece.

$$\mathrm{P}_{\mathrm{poss}}(\mathrm{G}|\mathrm{situation}) = \mathrm{P}_{\mathrm{poss}}(\mathrm{G}|\mathrm{ball},\ \mathrm{match\ state}) \tag{3}$$

It defines each possession as a sequence of states with the possible next state merely dependent on the current state, which follows the theorem of Markov Process. States mentioned in the above study include zones (show in Fig. 6) which denote the ball's location on the pitch and ending state as scoring a goal or losing possession. Unlike expected goal model that calculates the possibility of scoring goal when shooting at any given location on the pitch [24], EPV model calculates the probability the possession will develop into a goal from a given location on the football field. It takes the impact of previous game state into consideration when looking at current state, which makes it a better way to evaluate football game states than directly using expected goal model.

After the release of the original EPV model, several advanced models were built on top of this with more contextual information, such as the positions of players embedded.

Amongst these studies, the EPV model introduced by Fernández in 2019 is one of the most comprehensive [25]. The model consists of three independent sub-models for passes, shots and ball-drives and calculates the expected value of the possession given all the spatiotemporal information of players and the ball in specific possession of the game.

We use the original version of the EPV model in our research since it is applied to images (Pitch Control maps) which don't contain enough contextual information required to run an advanced EPV model.

As shown in Fig. 7, in our Possession Evaluation Model, each frame of the real sequence and generated sequence is converted to a scalar matrix showing the probabilities of the attacking team controlling the ball at given locations. We use the EPV algorithm introduced by Sarah in 2011 to generate the EPV Grid Map matrix, which denotes the expected possession value of the attacking team when given the location of the ball.

With this matrix, we can compute the EPV for each frame following Eq. (4).

$$EPV_{a_k} = \sum_{i=1}^{m}\sum_{j=1}^{n} EPV_{ij} * a_{t_k}^{ij} \tag{4}$$

As shown in Eq. (4), at a given location, we have the expected possession value, EPV_{ij} and the possibility of the attacking team controlling the ball, $a_{t_k}^{ij}$. The sum of the product of these two values represents the expected possession value for the Pitch Control map. This value indicates how likely the given game state shown as a Pitch Control map will develop into a goal. Therefore, a higher value suggests a better circumstance for the attacking team and a worse circumstance for the defending team.

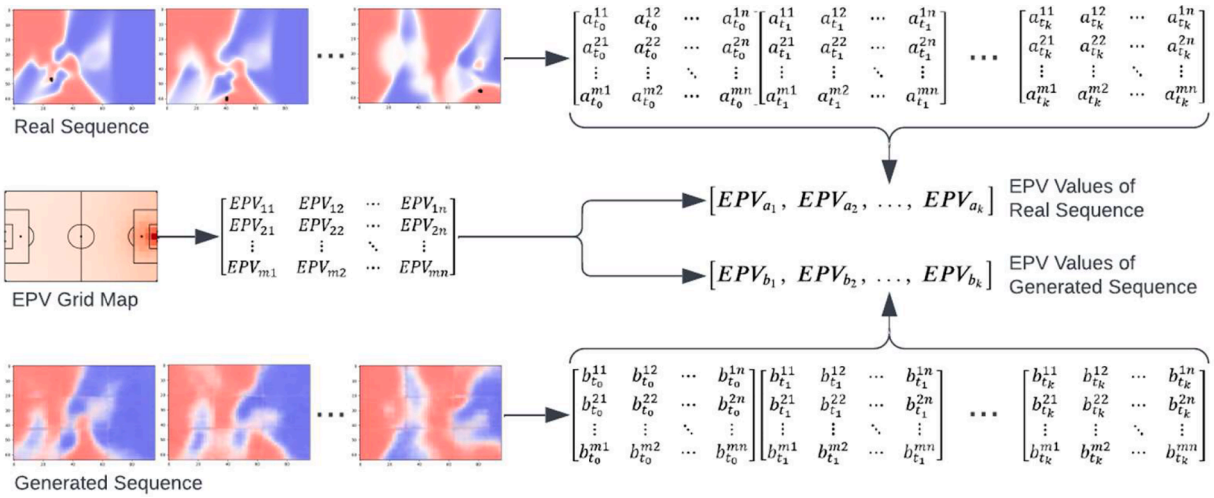


Fig. 7. Architecture of Possession Evaluation Model.

The EPV of the real sequence, EPV_a, and the expected possession value of the generated sequence, EPV_b are computed by summing the expected possession value for each frame respectively. EPV_a denotes the performance from both sides in real-life possession and EPV_b denotes the performance from both sides in the expected possession. The difference between EPV_a and EPV_b is used as the metric for quantifying the space-control efficiency and evaluating the team's attacking and defending performance. To be specific, a bigger EPV difference suggests a more efficient possession for the attacking team and a less efficient defending performance for the defending team. When evaluating the team performance in Section 4.1, the metric for evaluating defending efficiency is the negative value of the EPV difference.

Overall, the Possession Evaluation Model can quantify how efficiently one team controls the space by using the Space-control Prediction Model's output as the benchmark and calculating the difference between real possession and benchmark possession with EPV.

3. Results

To get the best predicted possession which represented the league-average performance, we train the Space-control Prediction Model and evaluate it by using the Structural Similarity Index Measure as an evaluation metric. To evaluate team performance in controlling spaces, we apply the Possession Evaluation Model by comparing the EPV of the Pitch Control map sequence for real-life possession with the EPV of the Pitch Control map sequence for predicted possession.

We conduct quantitative and qualitative experiments on Pitch Control map sequence dataset with our Space-control Prediction Model and it is proved that our model performs well. Then we discuss the results of Possession Evaluation Model.

3.1. Space-control prediction model

3.1.1. Evaluation metric

To evaluate the performance of our model on Pitch Control map sequence dataset, we need a suitable metric. When comparing two Pitch Control maps, the structural spatial difference instead of difference between each two pixels is the key factor. Therefore, the commonly used evaluation metric in image processing such as Cross Entropy can be inaccurate on our model.

Structural Similarity Index Measure (SSIM) is a method for measuring the similarity between two images [26]. It is a perception-based model which treats degradation as perceived change in structural information. It also incorporates crucial perceptual phenomena containing both luminance masking and contrast masking terms. Structural Similarity Index Measure consists of three components, l being the luminance, c being the contrast and s being the structure with three positive weights as α, β and γ [26].

The difference between SSIM and other techniques, such as Cross-Entropy and Mean Squared Error, which are used for quantifying the difference between two images, is that SSIM estimates the structural differences instead of absolute differences [27,28]. Structural differences denote that the pixels have strong inter-dependencies especially when they are spatially close. SSIM, other than Cross-Entropy and Mean Squared Error, is more suitable for calculating reconstruction loss when comparing Pitch Control maps since the structural spatial difference instead of the difference between each two pixels is the value aiming to be minimised. To be more specific, when comparing two Pitch Control maps which have lots of differences in details, they will still be viewed as a close pair if they show similar patterns which indicate the open space on the pitch. In the research conducted by Flynn et al., they discovered that SSIM was a superior objective image quality metric [29]. They applied SSIM to evaluate the difference between uncompressed and compressed images and invited human participants to do the evaluation at the same time. The results showed that when the SSIM value was over 0.95, a human couldn't perceive the difference between two images. It suggested that the SSIM metric can be effective in evaluating the difference between two images.

Table 1
Evaluation of Space-control Prediction Model in sequence reconstruction using SSIM measurement.

	Space-control Prediction Model	Vanilla VRNN Model
SSIM	0.954 ± 0.04 %	0.948 ± 0.10 %

Table 2
Evaluation of our Space-control Prediction Model in sequence prediction using SSIM measurement.

	Space-control Prediction Model	Vanilla VRNN Model
SSIM	0.850 ± 0.12 %	0.795 ± 0.25 %

3.1.2. Quantitative results

The SSIM metric is used to calculate the difference between real sequences and reconstructed sequences generated by our Space-control Prediction Model to evaluate the model performance. The results of the Vanilla VRNN Model are also shown for comparison. The model is evaluated in two ways. Firstly, the whole sequence and label for each frame (the first frame excluded) are fed to the models as input to get the output reconstructed sequence.

SSIM is used to evaluate the quality of each frame within the reconstructed sequence. As shown in Table 1, the SSIM of our Space-control Prediction Model is 0.954, over 0.95, which is the threshold for detecting the difference between two images by humans. It suggests that the sequence reconstructed by our model is good. Besides, the output of the Paired Sample T-test on the testing results of our model and the Vanilla VRNN Model is $P < 0.001$, which suggests the results of our model and the benchmark model has significant difference.

Secondly, only the first frame of the sequence and labels for each frame (the first frame excluded) are fed to the models as input to predict the following frames within the sequence. As shown in Table 2, the SSIM of the Space-control Prediction Model is significantly higher than the SSIM of the Vanilla VRNN Model, with 0.850 compared to 0.795. The SSIM value of sequence generation is lower than the sequence reconstruction with 0.850 compared to 0.954 since the model takes less information from the input sequence into account when generating the output. Besides, the output of the Paired Sample T-test on the testing results of our model and the Vanilla VRNN Model is $P < 0.001$, which suggests the results of our model and the benchmark model has significant difference. Specifically, the latent space of the model when reconstructing the sequence includes information about each frame (posterior) and the initial guess (prior), while the latent space of the model when predicting the sequence only includes the initial guess (prior). The predicted sequence shows what the Pitch Control map is most likely to be at each time-step when given the first frame and the labels. Since the model is trained on Premier League matches, the predicted sequence is believed to denote the league-average performance in terms of controlling the space on the pitch when given the first match scene of the possession and the types of movement of the controlled space (push, back or stay).

3.1.3. Qualitative results

From Fig. 8, we visualise the sequence of Pitch Control maps reconstructed by our Space-control Prediction Model and compare it with the real sequence and the sequence reconstructed by the Vanilla VRNN Model. The reconstructed frames show similar patterns with great image quality, which, together with the higher SSIM difference between the reconstructed frame and the real frame, has shown that our model performs well in reconstructing the Pitch Control maps within each possession sequence.

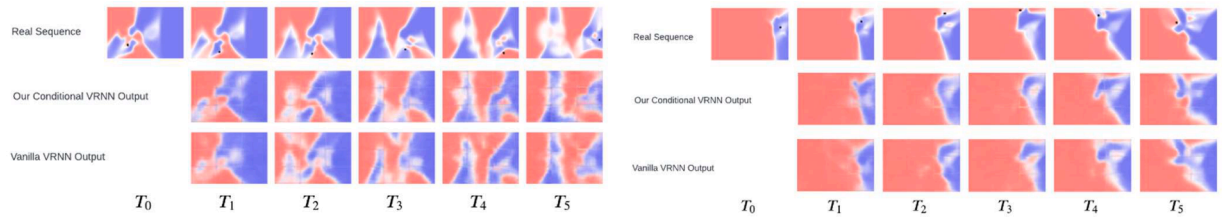


Fig. 8. Comparison of real sequence, sequence reconstructed by Vanilla VRNN Model and sequence reconstructed by our model.

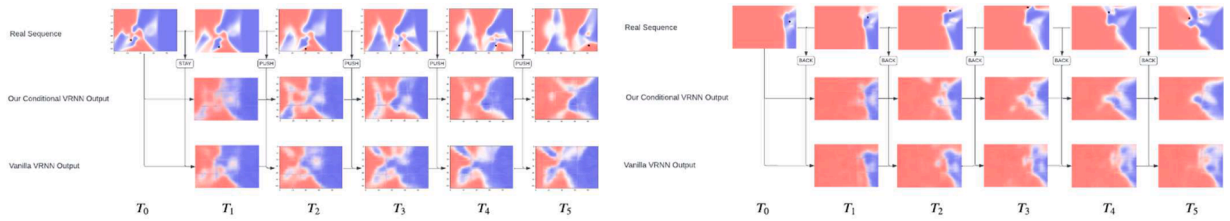


Fig. 9. Comparison of real sequence, sequence generated by Vanilla VRNN Model and sequence generated by our model.

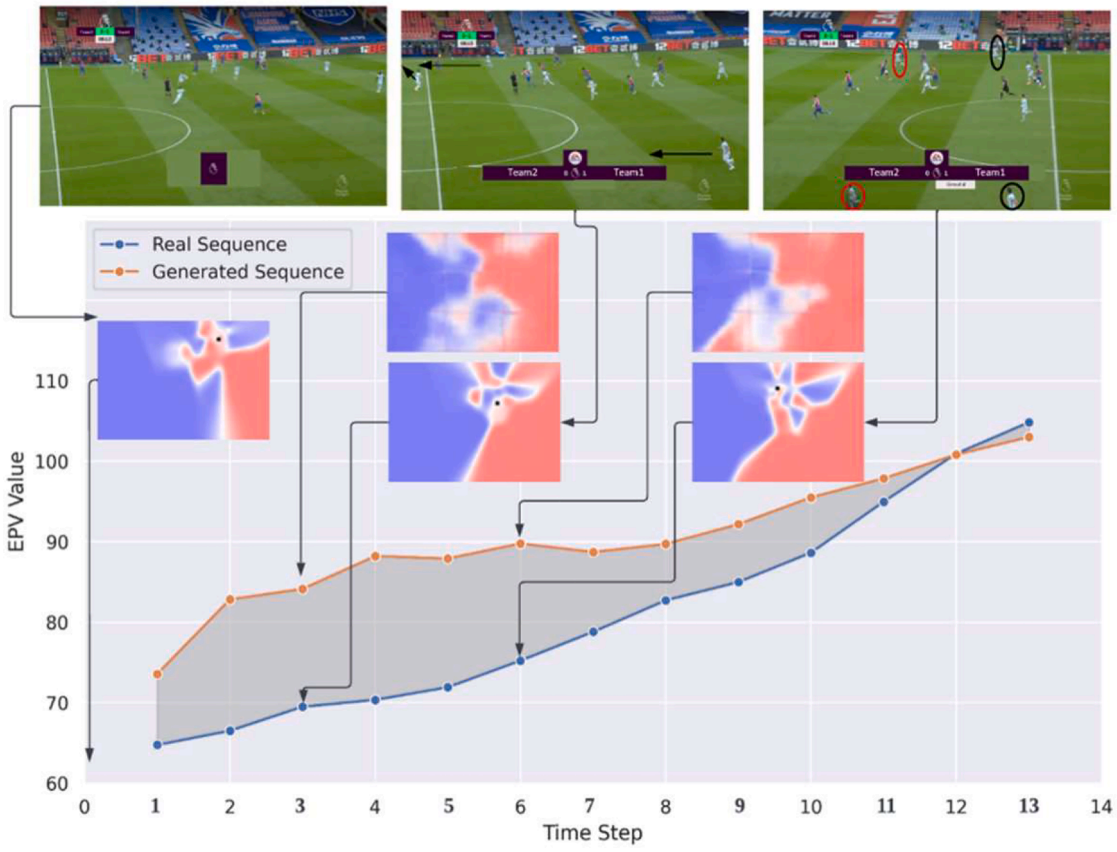


Fig. 10. The results of Possession Evaluation Model on one possession in the game between Team 2 (red kit) and Team 1 (white kit). Red area represents the area controlled by attacking side (Team 1 players wearing white kits). Blue area represents the area controlled by defending side (Team 2 players wearing red kits).

Moreover, to compare the real-life possession with predicted league-average possession effectively, we need to generate a sequence which has the same pattern (push, back or stay) as the original sequence. With only the first frame and labels for frames of the real-life sequence, we compare the sequence generated by our model with the real sequence and the sequence generated by the Vanilla VRNN Model to evaluate the effectiveness of our model in predicting league-average possession with the same pattern as real-life possession.

Fig. 9 shows the real sequence, the sequence generated by our model and the sequence generated by the Vanilla VRNN Model, only taking the first frame and labels as input.

At each timestep, the frames are reconstructed by conditional sampling in the later space of the VRNN Model with embedded conditions deciding the changes from the last frames. Compared to Vanilla VRNN, which only makes the good prediction at T1, our model predicts the pushing pattern of the sequence. Besides, from the left graph of Fig. 9,

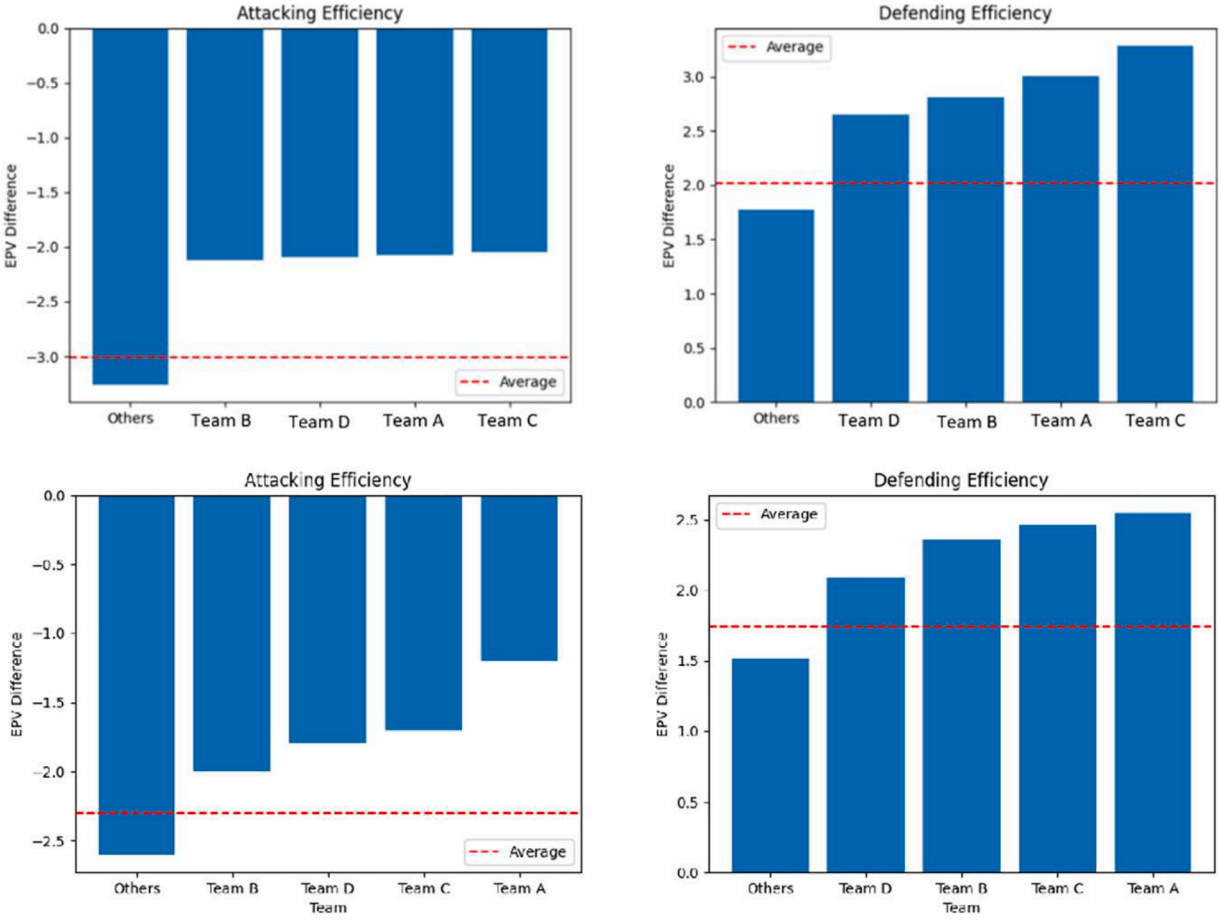


Fig. 11. Space-control efficiency in attacking and defending of top four Premier League teams and the others. Top: results on all data. Bottom: results on testing dataset.

our model shows a sharp pushing movement which precisely describes what happens in the real sequence. From the right graph of Fig. 9, when there is a backing trend in the real sequence, our model detects it and makes precise predictions to show this trend.

In general, our model performs well in reconstructing the real-life possession sequence and predicting the league-average sequence based on the first frame and labels for the frames based on the results of quantitative and qualitative experiments. More qualitative results are shown in the Appendix.

3.2. Possession evaluation model

With the Possession Evaluation Model, we can calculate the EPV difference between real sequence predicted league-average sequence and use this value to evaluate whether the specific team performance outperforms or underperforms the expectation. In this section, we illustrate why our Possession Evaluation Model is effective in evaluating team performance.

The Possession Evaluation Model is expected to evaluate team performance efficiently. We apply the Model to a possession in the game between Team 1 and Team 2 and prove the validity of our Model in making effective evaluations.

As shown in the first match frame from Fig. 10, this possession started with a Team 1 player gaining control of the ball in midfield while 7 Team 2 players were at the opponents' half pitch. It indicated that there could be a counterattack opportunity for Team 1, while this possession ended as a positional attack instead of a counter-attack in real life. As shown in the line chat, the EPV of the real sequence increased rapidly and reached above 100 while it was still lower than the EPV of the predicted sequence for the majority of the time. If using the EPV Model to evaluate this possession, the conclusion could be that this was a successful counter-attack with rising EPV. However, our Possession Evaluation Model suggested that this possession could be below the expectation.

By comparing the first and the second match scene in Fig. 10, Team 1 players didn't run forward quickly after gaining possession. According to the predicted control map at this point, either the Team 1 striker or the Team 1 right winger should have been much closer to the opponent's goal in the direction shown as arrows in the match scene. They were expected to utilise the potential open space in the right wing instead of staying still in real life. At this point, if using the EPV model to evaluate the possession, the only information that can be gained is that the attacking side's movement is positive with increasing EPV. Our model suggested that the counter-attack could have been more effective if Team 1 had gained the open space in the wing with the striker or winger running forward sooner.

At the 6th time-step, the EPV of the predicted Pitch Control map was higher than that of the real match Pitch Control map. In the predicted Pitch Control map, both the right wing and left wing were occupied by the attacking team, while in the real match Pitch Control map, the attacking side didn't utilise the right wing and occupied less space in the left wing near the opponent's goal. Team 1 players were making the counter-attack and pushing forward, as indicated by the rising EPV, whereas they could have been more efficient in terms of controlling more valuable spaces near the opponent's goal quickly after gaining possession. From the match scene at this time-step, black circles were the real positions of Team 1 wingers and red circles indicated the expected positions of Team 1 wingers according to the predicted Pitch

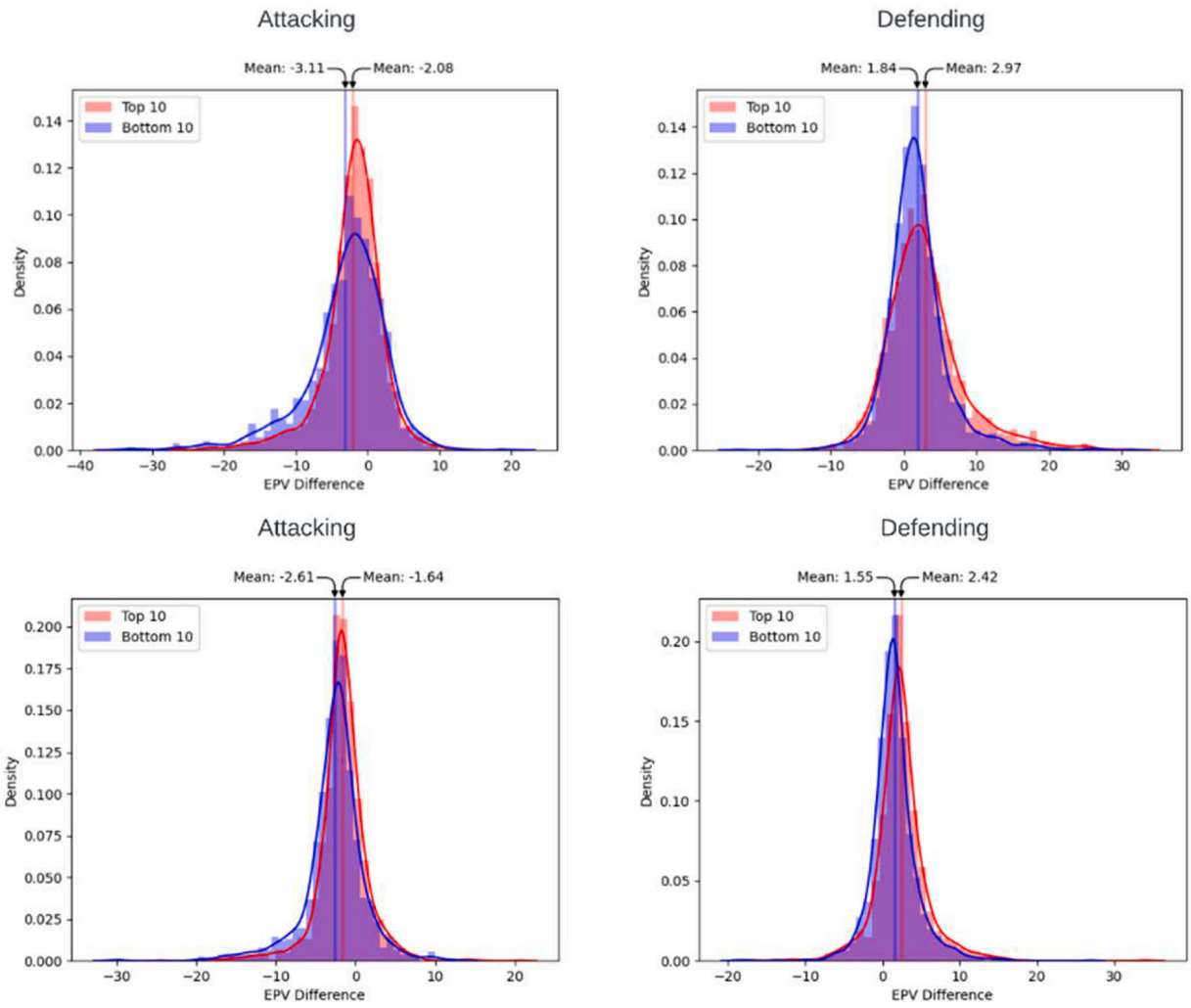


Fig. 12. Probability distribution of top 10 and bottom 10 teams' EPV Differences (output of Possession Evaluation Model) in attacking and defending. Top: results on all data. Bottom: results on testing dataset.

Control map. Together with the previous match scene, it could be argued that if Team 1 right winger had started running sooner and the left winger had run faster, Team 1 could meet the expectation and make a more efficient rapid counter-attack instead of being compelled to make a positional attack when opponents came back to defend.

In general, our Possession Evaluation Model is effective in evaluating team behaviours. It can evaluate the possession objectively and accordingly by comparing the real sequence with the predicted sequence. Even when the EPV of the real possession surges, whether the team outperforms the expectation in this possession will still be objectively evaluated by comparing it with the predicted possession through our Possession Evaluation Model.

4. Discussions

4.1. Analysis of team performance

While the data is collected from 19 to 20 Premier League season, we select the top four teams from that season, Team A, Team B, Team C and Team D, and check their attacking and defending efficiency using our Possession Evaluation Model. The model is run on all data from the 43 games and the data in the testing dataset, respectively. From Fig. 11, the averages of EPV differences of the possessions from the top four teams are computed, with the averages of other teams and all the possessions provided for comparison. The top four teams surpass the other teams on the dataset involving all the teams and the testing dataset with significant differences, which indicates the top teams create more valuable spaces on the pitch when they are controlling the ball. Moreover, the top four teams don't show apparent differences between each other on the dataset involving all the teams, and Team A outperforms the others on the testing dataset, which suggests they all have decent performance in terms of creating more open space in attacking while Team A is slightly better. Team A reaches the 1st place in the league table, which can be attributed to the great ability of utilising spaces according to the output of our model. At last, the top four teams outperform the other teams in defending efficiency on both datasets, which indicates they are more efficient in restricting the opponents' open area.

Team D is the worst amongst the top four on both datasets, as shown in the right charts, which suggests they are not performing well enough in defence. According to the Premier League table, they concede more goals than the other three, which is identical to the evaluation from our model.

From Fig. 12, the EPV difference of both top 10 and bottom 10 teams follow the normal distributions on both datasets. Besides, the EPV differences of the top 10 teams are higher than those of the bottom 10 teams on both datasets, which indicates that the top 10 teams create more valuable spaces on the pitch when attacking and allow the opponent to have less space when defending compared to the bottom 10 teams. In general, the results show that the teams' space-control abilities evaluated by our model are closely related to their performance in terms

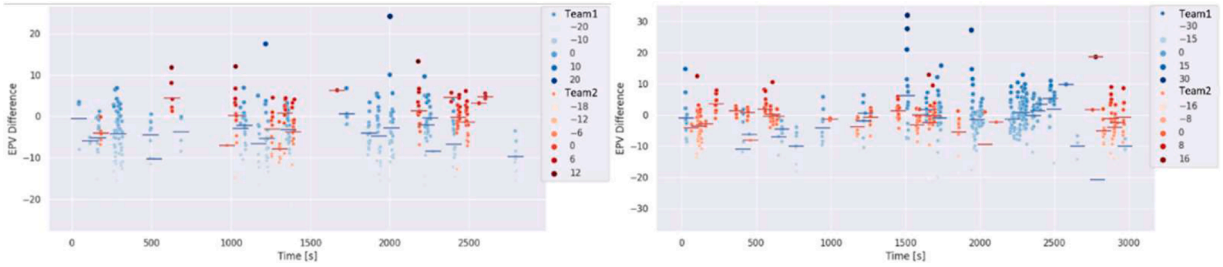


Fig. 13. Average of EPV differences (output of Possession Evaluation Model) between real and predicted Pitch Control maps within each possession in the first half (left) and the second half (right). The horizontal line shows the average of EPV differences for each possession.

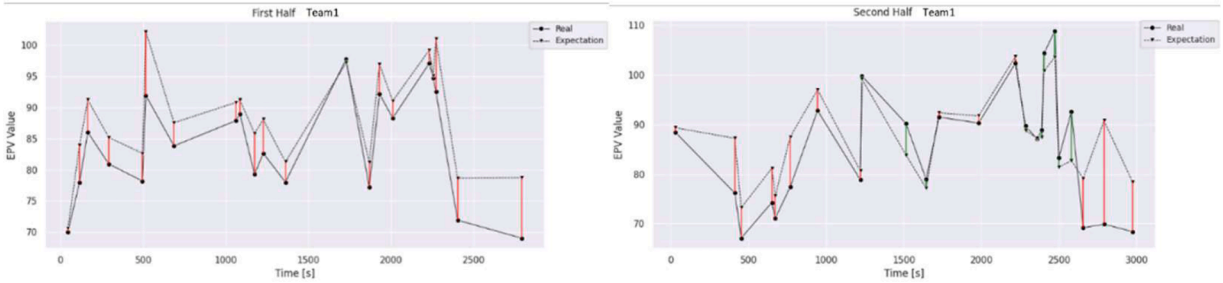


Fig. 14. The average EPV of real sequence and predicted sequence for each possession of Team 1 in the first half (left) and the second half (right). The vertical line shows the difference between EPV of real and predicted sequence with green denoting the real one outperforms the predicted one and red denoting the real one is worse than expectation based on prediction.

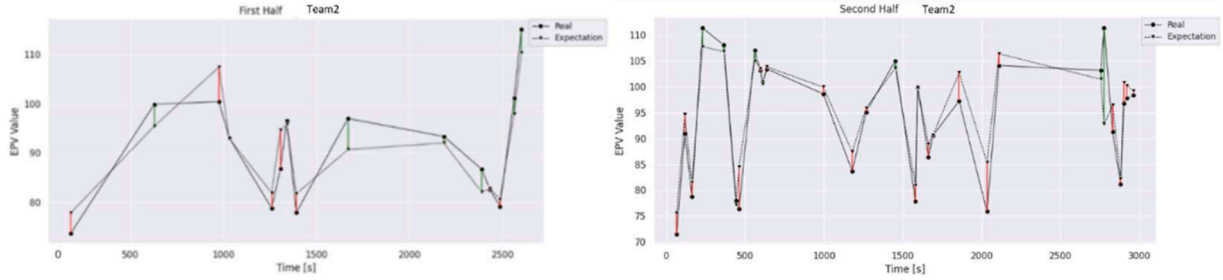


Fig. 15. The average EPV of real sequence and predicted sequence for each possession of Team 2 in the first half (left) and the second half (right). The vertical line shows the difference between EPV of real and predicted sequence, with green denoting that the real one outperforms the predicted one and red denoting that the real one is worse than expectation based on prediction.

of winning the match. It can be suggested that our model is an effective tool for the coaching staff to evaluate their own team's performance and the opponent's performance.

4.2. Analysis of the match between Team 1 and Team 2

Above is the overview of the teams' performance. Analysing one specific game between Team 2 and Team 1 with our Possession Evaluation Model provides more insights into tactics.

As shown in Fig. 13, Team 2 did better in the first half in terms of having more possessions outperforming the predicted possessions, while Team 1 outperformed the opponent in the second half. Team 1 was one of the top teams in the league, and Team 2 was one of those fighting to stay in the league. In the first half, Team 1 had a higher possession percentage, as suggested in the left graph with more dots in blue compared to red. In limited possessions from Team 2 in the first half, as shown in Fig. 15, they showed efficiency when attacking by outperforming the predicting sequences 6 times. Team 1 scored two goals, and Team 2 scored one goal in the first half. In the second half, the number of possessions from Team 2 increased according to the comparison of the number of red dots in the left and right graphs in Fig. 13. In the meantime, the quality of Team 1′s possessions showed a significant increase, with five more possessions outperforming the model prediction compared to the first half, as shown in Fig. 14. Since Team 2 was behind in the score sheet and wanted to score goals urgently in the second half, the team spread players around the pitch and played more build-up plays which could help to attack effectively [30].

However, this change could make the team vulnerable when they lose possession [31,32]. The spread formation made it harder for them to keep compact form when coming back to defence, which gave Team 1 more chances to create valuable spaces and make efficient attacks. In conclusion, our model can be a useful tool to assist the coaching staff to evaluate the performance from their team and the opponent team.

4.3. Analysis of a successful counterattack possession

In the second half, Team 1 scored their winning goal in the 1519th second. According to the match footage, a successful counterattack led to this goal. We analysed this possession comparing the Pitch Control maps generated by our model and the real sequence. As shown in Fig. 16, this possession started with a Team 1 player winning an aerial challenge. At this moment, Team 2 players were pushing high upfront

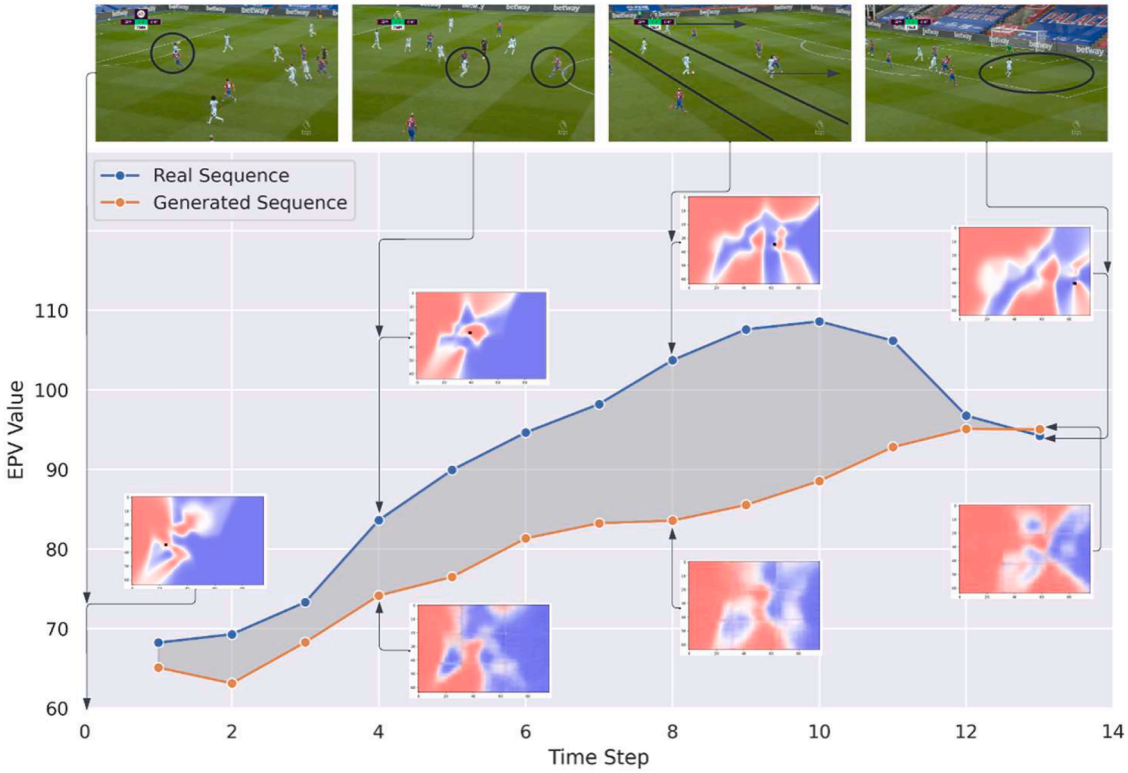


Fig. 16. Decomposing the successful counterattack by Team 1 in the second half. Interval between two time-steps is 1 second. Red area represents the area controlled by attacking side (Team 1 players wearing white kits). Blue area represents the area controlled by defending side (Team 2 players wearing red kits).

the pitch as shown in the Pitch Control map's blue area at the beginning. At the fourth timestep, one of the Team 1 midfielders was about to get the ball and started the counterattack and the closest Team 2 player decided to run to the opponent and stopped the counterattack from the beginning. To this point, the EPV of real sequence remained close to the one predicted by our model which suggested the situation was like other counterattacks happening in the league and had limited chance of scoring a goal. However, after several frames, the EPV of Team 1's possession rapidly increased and surpassed prediction much further. At the 8th time-step, the blue defensive area was cut off in the middle with the crucial area occupied by the opponent in real possession while the defensive side kept the compactness in the predicted possession, which directly led to the big difference between real frame and predicted frame EPV values. The surging difference of EPV indicated an increasing chance of making threats to the opponent by Team 1. The reason why Team 1 suddenly dominated the valuable area was that the Team 2 player failed to stop the Team 1 midfielder and left his behind totally open for the opponent to run the counterattack. The Team 2 defenders were running backwards to cover the space in front of the goal while the other players didn't manage to back at the same pace and the defensive formation of Team 2 was disjointed as shown in the Pitch Control map and game footage frame at 8th timestep. From the match frame at this moment, the Team 1 ball carrier directly faced the defenders, and left and right-wing areas were going to be controlled by two running-forward strikers as shown in the Pitch Control map, which indicated that the defensive side was in a passive situation [33].

This directly led to the defensive side losing control of space close to the goal as shown in the Pitch Control map, which made them vulnerable and finally concede a goal. In the last two time-steps, the EPV of the real sequence decreased since the defenders covered the open space in the right back and reduced the overall open space on the pitch. However, this retrieval was too late for Team 2 when the Team 1 striker already had a wide-open chance for scoring.

When looking back at this whole possession with the above analysis enabled by our model, it can be argued that the cause of the failure from the defensive side is the bad coordination of players. Specifically, when Team 1 was about to launch the counterattack at the 4th timestep, there were options for the closest Team 2 player, making an aggressive tackle or running back to help defenders [34]. If this player chose the first option, just as what happened in the real game, the defenders should be aggressive at the same time, trying to stop the opponent at the first place or force the opponent to fall into an offside trap. However, the defenders chose to go backwards since the above option could be too risky [35]. If that Team 2 player chose the second option, which was running backwards, it would be most likely that their controlled area would not be cut off by the opponent, and the defenders would not have to face the opponent alone without any help. From the perspective of the attacking side, there were two keys to the successful counterattack. At first, in this possession, the great run of the left winger created lots of space in the wing which made the EPV of real possession outperform the predicted possession. Secondly, at the 8th time-step, when the Team 1 midfielder held the ball, the Team 1 striker made a great run utilising the potential space in the right wing, which enabled him to score the goal at the end of this possession.

In general, with our model, the coaching staff can have a direct sense of how their team performs in a specific possession by looking at the graph showing the differences in space control and the efficiency in attacking and defending between the real possession and the benchmark possession. Our model can be a useful addition to existing analysing tools for coaching staff to analyse the game more efficiently.

4.4. Analysis of individual performance

Our model can not only be applied to evaluate team performance at the team level but be used to make individual assessments as well. To evaluate the on-ball contribution of football players, the best way is to look at their performance in passing and dribbling [36]. For each successful passing and dribble event in our dataset, the Pitch Control maps

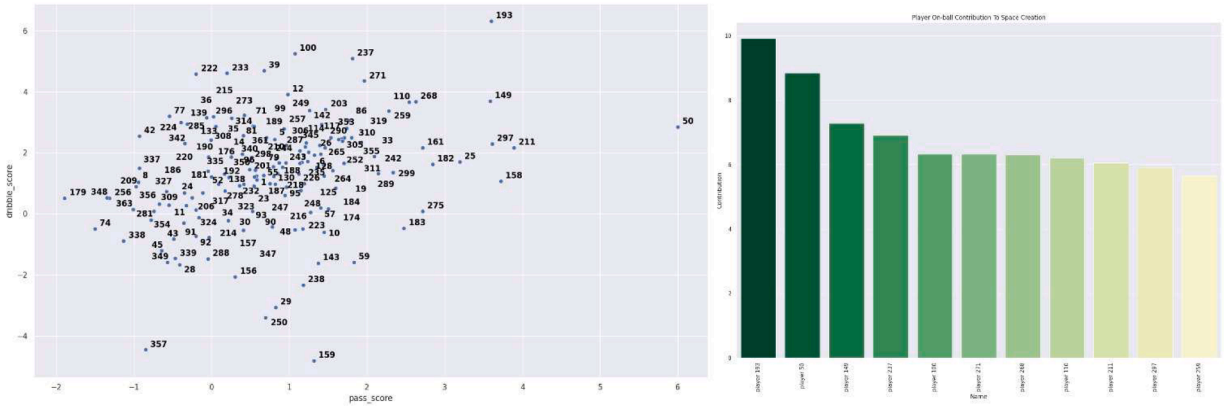


Fig. 17. Individual performance on all data. Left: individual average contribution per passing event and dribbling event; right: individual average contribution per on-ball event (passing or dribbling). The numbers in the scatter graph denote the players' IDs.

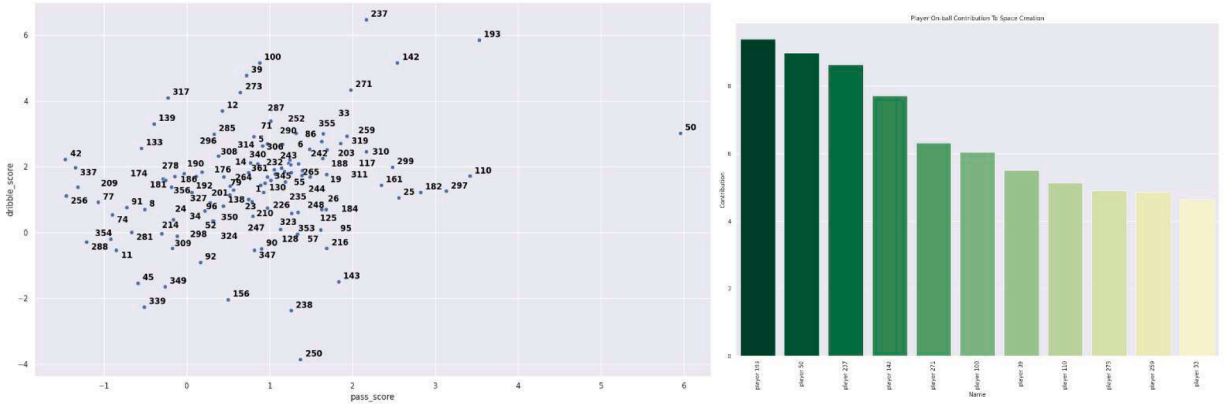


Fig. 18. Individual performance on testing data. Left: individual average contribution per passing event and dribbling event; right: individual average contribution per on-ball event (passing or dribbling). The numbers in the scatter graph denote the players' IDs.

within this event are processed by our Possession Evaluation Model. The model output is used to quantify the contribution to space creation by the player who executes the event. All players' names are replaced by indexing to avoid the disclosure of confidential information. From Fig. 17, on the dataset involving all the possessions, Player 50, a midfielder surpasses all the others in terms of passing contribution to space creation, while Player 193, also a midfielder, is the most efficient dribbler considering creating space for the team. According to Fig. 18, the result on testing dataset shows the similar pattern with Player 50 reaches top in the passing contribution and Player 193 is the second most efficient dribbler. Moreover, as shown in Fig. 17 and Fig. 18, Player 193 and Player 50 are the top two players on both datasets, regarding the on-ball contribution to attacking in space creation. Overall, our model can be used as an assistant tool for coaches to evaluate the performance of players in the team and for the scouts to evaluate the space-creation abilities of players who are about to be recruited.

5. Conclusion

The objective of this paper is to build a novel Machine Learning framework to quantify football team space-control efficiency and evaluate team performance based on that. To achieve this, we construct a Space-control Prediction Model to predict league-average performance in space control and a Possession Evaluation Model to compare the predicted possession sequence and real possession sequence for evaluating team efficiency in controlling the open space during the match.

According to the results of quantitative and qualitative analysis, the Space-control Prediction Model is proven to be capable of making stable long-term predictions of Pitch Control maps when given the starting frame of the possession and the label for each frame. The embedding of domain knowledge, the pattern between every two adjacent frames, enables the model to make more precise predictions conditioning on contextual factors. Besides, the Possession Evaluation Model is proved to be more consistent and objective compared to the EPV Model when quantifying team space-control efficiency. With the output of the Space-control Prediction Model as the benchmark, the team's and player's performance can be evaluated more accurately. With our Possession Evaluation Model, the coaching staff can make decisions more efficiently based on the evaluation of a team's and individual's performance. The remarkable performance of the model is ascribed to the implementation of deep generative modelling on ample game possession sequences covering 43 games in the high-level professional football league.

With our proposed framework, a coach can evaluate the attacking and defending efficiency of the team within a period from the whole season to individual games. For a specific game, our model can evaluate each team's performance in controlling valuable areas on the pitch within this game. The coach can quickly locate the possessions where the team outperforms the expectation or does not meet the expectation. For possession showing a significant difference from that predicted by our model, the coach can look into this sequence and analyse the decision-making of each individual by comparing the real frame with the predicted frame at each timestep so that a pointed adjustment can be made on that. Moreover, the coaching staff can also evaluate individuals' contributions to team space creation utilising our framework.

Future work will focus on using our Possession Evaluation model to

analyse different types of situations in football games. After classifying possessions, we can use our model to evaluate the team performance in different categorised possessions, such as counterattack and build-up play, instead of analysing all possessions at once. The specific insights of tactics are expected to be provided on given scenarios by the future approach. Furthermore, in this research, due to sensitivity of the data, we only have limited amount of data for training our model. Although we have ample data to achieve ideal results in this research, the results can have even stronger validity for the coaching staff with more data to train our model in the future. Since we don't have connections to the coach at the moment, we didn't invite coaches to validate our model's effectiveness in real-life applications. Involving the coaches' validation in the future will bring more validity to this research.

Author agreement

All authors agree to submit the paper to your journal and have future publications.

CRediT authorship contribution statement

Chaoyi Gu: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Varuna De Silva:** Conceptualization, Writing – review & editing, Supervision, Project administration. **Mike Caine:** Writing – review & editing.

Declaration of Competing Interest

The data used in this research is confidential and cannot be shared with the public.

Data availability

The authors do not have permission to share data.

Funding

This research is partly funded by the Engineering and Physical Sciences Research Council, under grant number: EP/T000783/1: MIMIc: Multimodal Imitation Learning in Multi-Agent Environments.

Appendices

A. *Technical details of space-control prediction model*

To predict the possession when given only the first frame, we build a generative model which can predict the sequence conditioned on the first frame and the condition for frame at each times-step.

Vanilla VRNN model can generate sequences by randomly sampling the latent space. However, the randomly generated frames cannot be controlled and used for meaningful inference. Therefore, we propose a Conditional VRNN architecture with pre-trained CNN-LSTM Classifier embedded to generate the sequence conditioning on the same labels from real-life sequence. The generated sequence represents the likely performance in this possession.

The overall structure of the model is shown in Fig. 7. X_t denotes the pitch frame at time t, h_t denotes the hidden state of LSTM networks at time t and z_t represents the latent space at time t. We use  to represent the merging layers,  to represent the splitting layers, and  to represent the neural networks.

Frame encoder

The encoder is built with stride convolutions and Rectified Linear Units (ReLu) which extract latent representations from input frame at each timestep of every sequence.

Prior and approximate posterior

The prior at time t is ψ_p^t which is parametrised as a diagonal normal distribution $\mathcal{N}(\mu_{0,t}, diag(\sigma_{0,t}^2))$ where $\mu_{0,t}$ and $\sigma_{0,t}$ are extracted from h_{t-1}, the recurrent hidden state at time t-1. The approximate posterior at time t is ψ_q^t which is parametrised as another diagonal normal distribution $\mathcal{N}(\mu_{z,t}, diag(\sigma_{z,t}^2))$ where $\mu_{z,t}$ and $\sigma_{z,t}$ are extracted from the concatenation of latent variables encoded from input frames and h_{t-1}. The posterior at time t will be used to inference getting z_t which can be used to reconstruct the input image.

Frame decoder

At each timestep t, the decoder takes the representation of the previously generated frame x_{t-1} and variables z_t sampled from the posterior to generate x_t based on $p(x_t|z_t, x_{<t}.c)$. The decoder consists of transposed convolutions and Rectified Linear Units (ReLu).

Recurrence

To capture the temporal information amongst frames within a video, the LSTM network is applied to denote the recurrent relationship amongst frames and help generate consistent sequences. The hidden state h_t at time t is updated according to encoded input frame x_t, latent variables z_t, reshaped conditional variables c_t and hidden state h_{t-1} at previous timestep. The initial hidden state is set to zero vector.

Conditioning

The conditions are acquired through feeding the two contiguous frames to pre-trained classifier which outputs the vector $[P_1, P_2]$. P_1 and P_2 representing the possibility of the short sequence being pushing and backing respectively. While P_1 and P_2 are both less than 0.9, the prediction of label is viewed as ambiguous, and the sequence is labelled as staying. Then the generated labels are processed by one-hot encoding before implemented into VRNN. Three approaches are made to implement conditions into VRNN model.

At first, the prior is conditioned on the generated label. The prior therefore is conditioned on not only the LSTM hidden state vector $h^{(t-1)}$ but condition vector a^t as well. The prior distribution follows the distribution.

$$\psi_p^t = \left[\mu_{0,t}, \sigma_{0,t}\right] = \psi_\tau\left(\psi_\tau\left(h^{(t-1)}\right), a^t\right) \tag{A1}$$

Secondly, the posterior is conditioned on the encoded input frame, condition vector a^t and LSTM hidden state vector at previous timestep $h^{(t-1)}$. The posterior distribution follows the distribution.

$$\begin{aligned} \psi_q^t = \left[\mu_{z,t}, \sigma_{z,t}\right] &= \psi_\tau(\psi_\tau(\psi_\tau(\psi_{enc}(x^t)), a^t), \psi_\tau\left(h^{(t-1)}\right)) \\ \psi_{q_1}^t &= \psi_{enc}(x^t) \\ \psi_{q_2}^t &= \psi_\tau(\psi_\tau(\psi_{enc}(x^t)), a^t) \end{aligned} \tag{A2}$$

ψ_τ denotes the neural networks containing Fully-Connected and ReLu layers and ψ_{enc} denotes the encoder.

At last, the parameters of LSTM networks are conditioned on the generated label. Three groups of parameters correspond to three different labels.

Model learning

Loss function contains the difference between posterior and prior (KL divergence) and reconstruction loss (Structural Similarity Index Measure). The loss is calculated at each frame and then the mean of losses across all frames is calculated.

$$\mathbb{E}_{q(z\le T|x\le T)}\left[\sum_{t=1}^{T}(-KL(q(z_t|x_{\le t}, z_{<t})\|\, p(z_t|x_{<t}, z_{<t})) + logp(x_t|z_{\le t}, x_{<t}))\right] \tag{A3}$$

B. Technical details of evaluation metric (SSIM)

Structural Similarity Index Measure (SSIM) is a method for measuring the similarity between two images (Wang et al., 2004). It is a perception-based model which treats degradation as perceived change in structural information. It also incorporates crucial perceptual phenomena containing both luminance masking and contrast masking terms. As shown in Eq. (B1), Structural Similarity Index Measure consists of three components, l being the luminance, c being the contrast and s being the structure with three positive weights as α, β and γ.

$$\mathrm{SSIM}_{(x,y)} = [l(x,y)]^\alpha * [c(x,y)]^\beta * [s(x,y)]^\gamma \tag{B1}$$

C. More Visualization Results of Space-control Prediction Model

Fig. C1 shows the real sequences, and sequences reconstructed by our Conditional VRNN model and Vanilla VRNN model for three different possessions with all frames within the sequence and the labels for all frames except the frames at T_0 as input.

Fig. C2 shows the real sequences, and sequences predicted by our Conditional VRNN model and Vanilla VRNN model for three different possessions with only the frames at T_0 and the labels for all frames except the frames at T_0 as input.

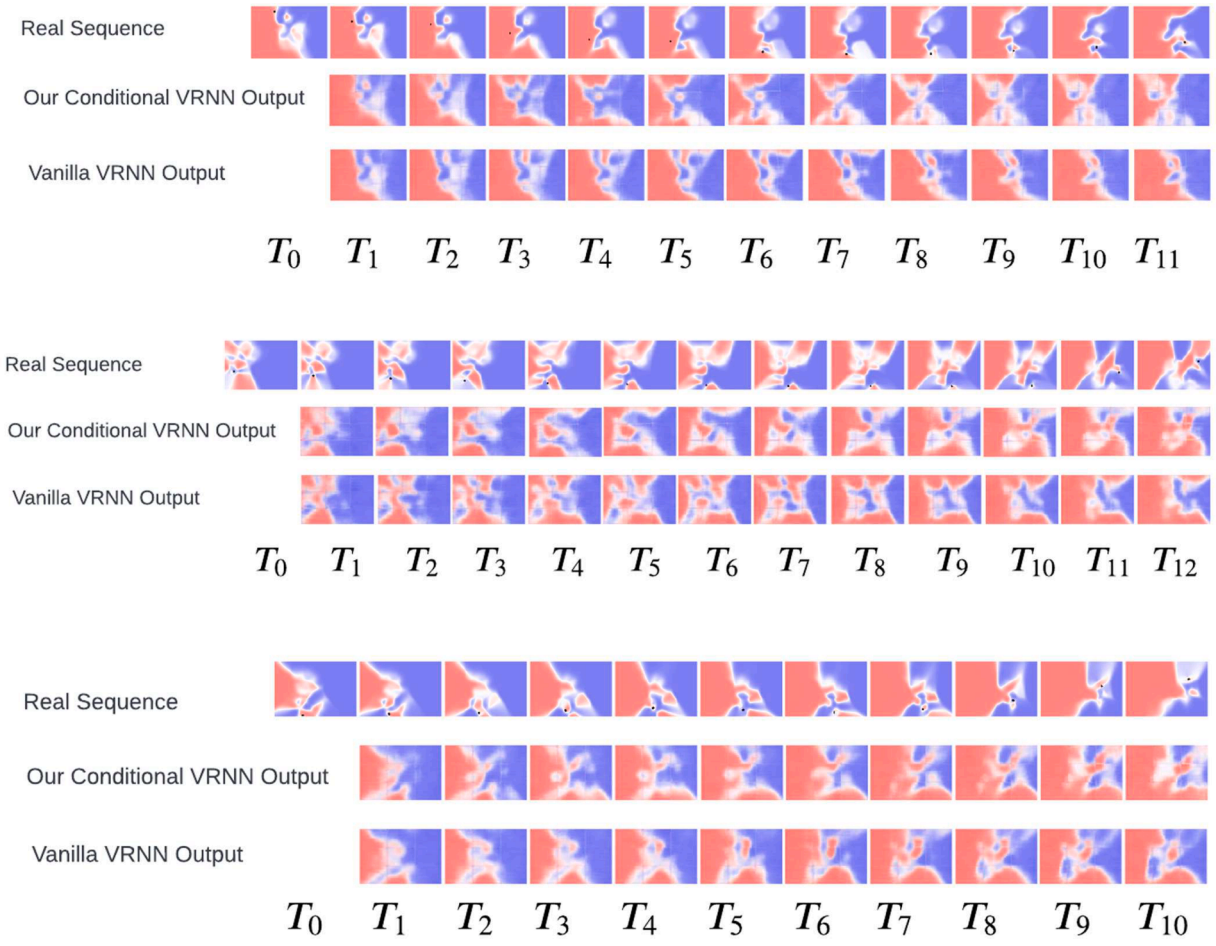


Fig. C1. Real sequences and sequences reconstructed by our model and Vanilla VRNN model on three different possessions.

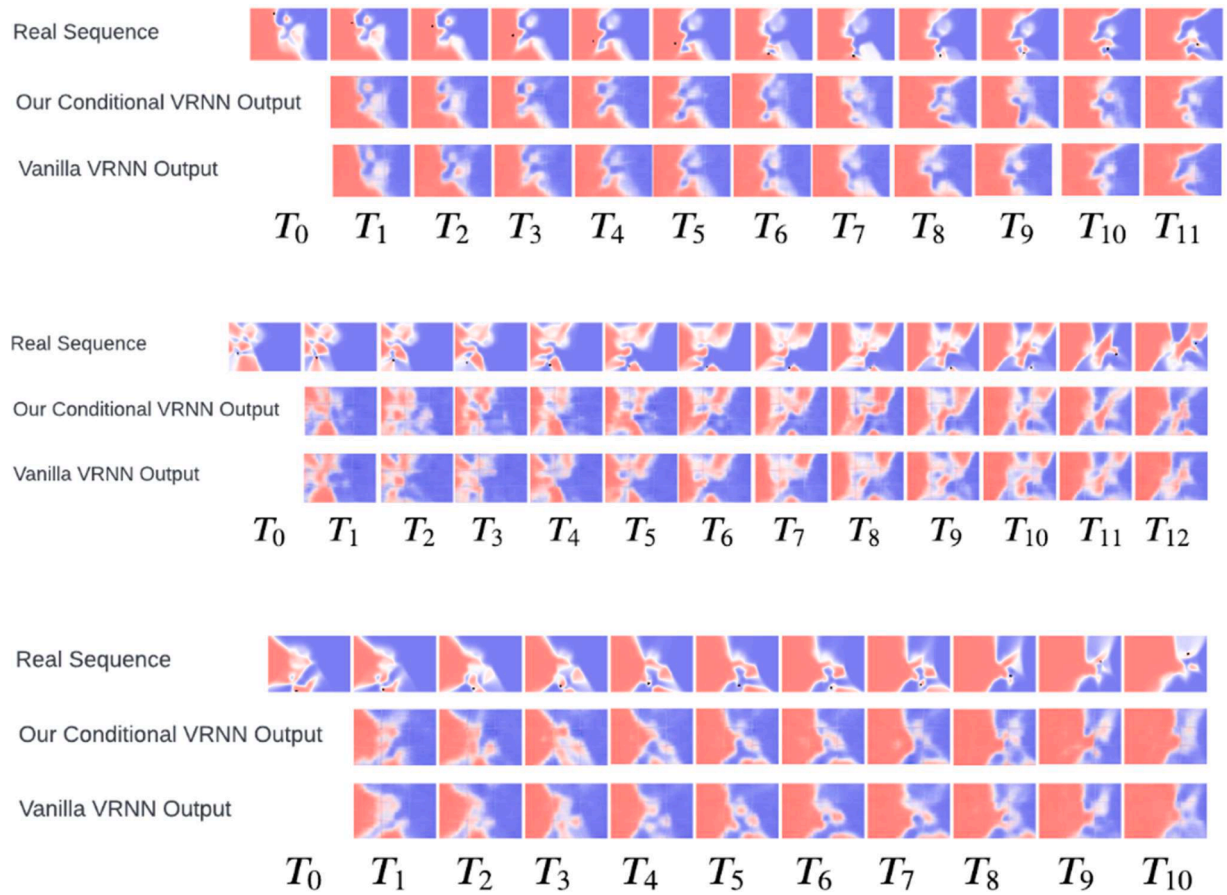


Fig. C2. Real sequences and sequences predicted by our model and Vanilla VRNN model with only frames at T_0 and labels as input on three different possessions.

References

[1] R. Mackenzie, C. Cushion, Performance analysis in football: a critical review and implications for future research, J. Sports Sci. 31 (6) (2013) 639–676.

[2] B. Min, J. Kim, C. Choe, H. Eom, R.B. McKay, A compound framework for sports results prediction: a football case study, Knowl. Based Syst. 21 (7) (2008) 551–562.

[3] A.C. Constantinou, N.E. Fenton, M. Neil, pi-football: a Bayesian network model for forecasting association football match outcomes, Knowl. Based Syst. 36 (2012) 322–339.

[4] A. Constantinou, N. Fenton, Towards smart-data: improving predictive accuracy in long-term football team performance, Knowl. Based Syst. 124 (2017) 93–104.

[5] R. Ievoli, L. Palazzo, G. Ragozini, On the use of passing network indicators to predict football outcomes, Knowl. Based Syst. 222 (2021), 106997.

[6] R. Baboota, H. Kaur, Predictive analysis and modelling football results using machine learning approach for English premier league, Int. J. Forecast. 35 (2) (2019) 741–755.

[7] G. Anzer, P. Bauer, A goal scoring probability model for shots based on synchronized positional and event data in football (soccer), Front. Sports Active Living 3 (2021), 624475.

[8] A. Arbues-Sanguesa, A. Martin, J. Fernández, C. Ballester, G. Haro, Using player's body-orientation to model pass feasibility in soccer, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, 2020, pp. 886–887.

[9] M. Stein, D. Seebacher, R. Marcelino, T. Schreck, M. Grossniklaus, D.A. Keim, H. Janetzko, Where to go: computational and visual what-if analyses in soccer, J. Sports Sci. 37 (24) (2019) 2774–2782.

[10] J. Fernández, L. Bornn, Wide open spaces: a statistical technique for measuring space creation in professional soccer, in: Sloan Sports Analytics Conference 2018, 2018.

[11] R. Rein, D. Raabe, D. Memmert, Which pass is better?" Novel approaches to assess passing effectiveness in elite soccer, Hum. Mov. Sci. 55 (2017) 172–181.

[12] S. Kim, Voronoi analysis of a soccer game, Nonlinear Anal.: Model. Control 9 (3) (2004) 233–240.

[13] G. Andrienko, N. Andrienko, G. Budziak, J. Dykes, G. Fuchs, T. von Landesberger, H. Weber, Visual analysis of pressure in football, Data Min. Knowl. Discov. 31 (6) (2017) 1793–1839.

[14] W. Spearman, A. Basye, G. Dick, R. Hotovy, P. Pop, Physics-based modeling of pass probabilities in soccer, in: Proceeding of the 11th MIT Sloan Sports Analytics Conference, 2017.

[15] S. Rudd. A framework for tactical analysis and individual offensive production assessment in soccer using Markov chains. In New England Symposium on Statistics in Sports, 2011. URL http://nessis.org/nessis11/rudd.pdf.

[16] S. Llana, P. Madrero, J. Fernández, F.C. Barcelona, The right place at the right time: advanced off-ball metrics for exploiting an opponent's spatial weaknesses in soccer, in: Proceedings of the 14th MIT Sloan Sports Analytics Conference, 2020.

[17] T.M. Mitchell, The role of unlabelled data in supervised learning. Language, Knowledge, and Representation, Springer, Dordrecht, 2004, pp. 103–111.

[18] J. Gu, Z. Wang, J. Kuen, L. Ma, A. Shahroudy, B. Shuai, T. Chen, Recent advances in convolutional neural networks, Pattern Recognit. 77 (2018) 354–377.

[19] J. Cheng, L. Dong, M. Lapata, Long short-term memory-networks for machine reading. Conference on Empirical Methods in Natural Language Processing, (2016) pp. 551–561.

[20] K. Greff, R.K. Srivastava, J. Koutník, B.R. Steunebrink, J. Schmidhuber, LSTM: a search space odyssey, IEEE Trans. Neural Netw. Learn Syst. 28 (10) (2016) 2222–2232.

[21] D.P. Kingma, M. Welling, An introduction to variational autoencoders. Foundations and Trends in Machine Learning, 12 (4) (2019) 307–392.

[22] D.P. Kingma, M. Welling, Auto-encoding variational bayes. International Conference on Learning Representations, (2013) abs/1312.6114.

[23] J. Chung, K. Kastner, L. Dinh, K. Goel, A.C. Courville, Y. Bengio, A recurrent latent variable model for sequential data, Adv. Neural Inf. Process. Syst. 28 (2015).

[24] A. Rathke, An examination of expected goals and shot efficiency in soccer, J. Hum. Sport Exerc. 12 (2) (2017) 514–529.

[25] J. Fernández, L. Bornn, D. Cervone, Decomposing the immeasurable sport: a deep learning expected possession value framework for soccer, in: 13th MIT Sloan Sports Analytics Conference, 2019.

[26] Z. Wang, A.C. Bovik, H.R. Sheikh, E.P. Simoncelli, Image quality assessment: from error visibility to structural similarity, IEEE Trans. Image Process. 13 (4) (2004) 600–612.

[27] D.M. Kline, V.L. Berardi, Revisiting squared-error and cross-entropy functions for training neural network classifiers, Neural Comput. Appl. 14 (4) (2005) 310–318.

[28] U. Sara, M. Akter, M.S. Uddin, Image quality assessment through FSIM, SSIM, MSE and PSNR—a comparative study, J. Comput. Commun. 7 (3) (2019) 8–18.

[29] J.R. Flynn, S. Ward, J. Abich, D. Poole, Image quality assessment using the ssim and the just noticeable difference paradigm, in: International Conference on Engineering Psychology and Cognitive Ergonomics, Berlin, Heidelberg, Springer, 2013, pp. 23–30.

[30] P.S. Bradley, C. Carling, D. Archer, J. Roberts, A. Dodds, M. Di Mascio, P. Krustrup, The effect of playing formation on high-intensity running and technical profiles in English Fa premier league soccer matches, J. Sports Sci. 29 (8) (2011) 821–830.

[31] N. Witt, Defensive compactness as a performance indicator for game annotation, Res. Q. Exerc. Sport 87 (S1) (2016) S59.
[32] D. Memmert, K.A. Lemmink, J. Sampaio, Current approaches to tactical performance analyses in soccer using position data, Sports Med. 47 (1) (2017) 1–10.
[33] V. Armatas, D. Ampatis, A. Giannakos, P. Sileloglou, Analysis of the Successful Counter-Attacks in High-Standard Soccer Games (No. RefW-40-7804), Aristotle University of Thessaloniki, 2005.
[34] H. Sarmento, M.T. Anguera, A. Pereira, A. Marques, J. Campaniço, J. Leitão, Patterns of play in the counterattack of elite football teams-A mixed method approach, Int. J. Perform. Anal. Sport 14 (2) (2014) 411–427.
[35] S. Takahashi, K. Yoshida, T. Kwon, K.H. Lee, J. Lee, S.Y. Shin, Spectral-based group formation control. Computer Graphics Forum, Blackwell Publishing Ltd, Oxford, UK, 2009, pp. 639–648. Vol. 28, No. 2.
[36] T.B. McGuckian, M.H. Cole, G. Jordet, D. Chalkley, G.J. Pepping, Don't turn blind! The relationship between exploration before ball possession and on-ball performance in association football, Front. Psychol. 9 (2018) 2520.